

PROJECT DESIGN PHASE – SOLUTION ARCHITECTURE

Solution Architecture Template

DATE	04/10/2026
TEAM ID	SWTID-2026-8904
PROJECT NAME	AI Study Buddy

MAXIMUM MARKS: 2 Marks

Solution Architecture

STUDENT	→	WEB / STUDY DASHBOARD	→	BACKEND (NODE.JS + EXPRESS.JS)
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				MONGODB GOOGLE GEMINI AI

Technology Stack:

- Node.js
- Express.js
- MongoDB
- Mongoose
- JWT
- bcryptjs
- Google Gemini AI

Explanation

- 1. User Layer:** Students interact with the study dashboard to learn topics, manage notes, take quizzes and ask questions.
- 2. Backend:** Node.js and Express.js handle authentication, study materials, quiz operations and AI integration.
- 3. AI Service:** Google Gemini generates explanations, summaries, examples and quiz content.
- 4. Database:** MongoDB stores users, notes, quiz data and progress history.
- 5. Communication:** Components communicate through REST APIs; HTTPS should be used for deployed environments.